Integral Invariant Discovery via Bayesian Experimental Design

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In collaboration with

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Work in progress — preliminary results

Invariant discovery: problem statement

Which candidate integrals remain approximately constant as the simulation evolves?

Given: A simulated MHD trajectory and a fixed bank of candidate integrals.

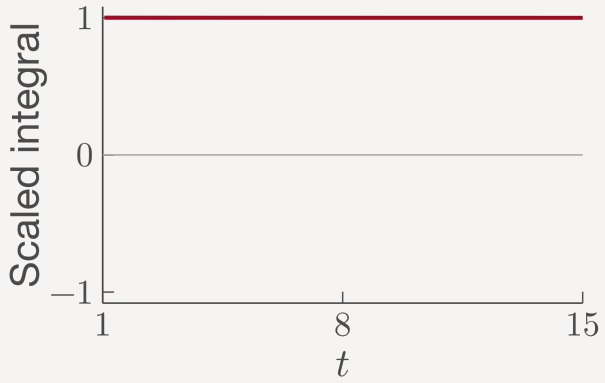
Goal: Classify each candidate as invariant or not invariant.

Problem: Evaluating every candidate at every spatial point and every saved time is expensive. We want to classify candidates by reading only a subset of points/times.

Examples of candidate integrals

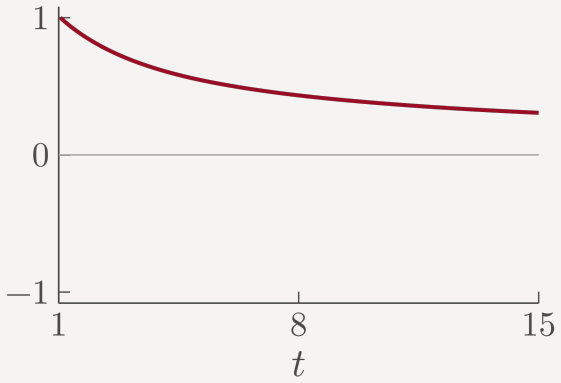
Approximately invariant

$$\int_\Omega \rho \, d^3x$$



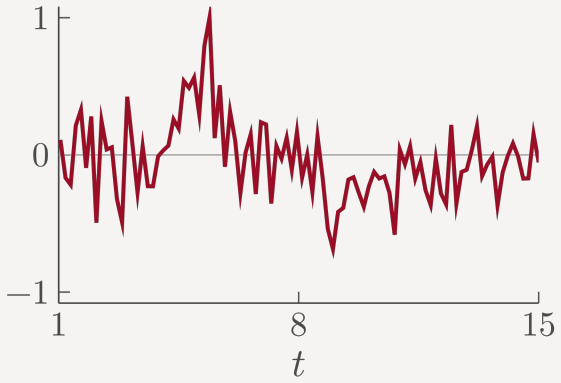
Not invariant (trend)

$$\int_\Omega |\mathbf{B}|^2 \, d^3x$$



Not invariant (fluctuating)

$$\int_\Omega \rho^2 \, \nabla \cdot \mathbf{A} \, d^3x$$



Candidate bank: generated from a bounded grammar, within set complexity limits

Number of field factors

$\rho^2, \quad \rho^3, \quad \rho|\mathbf{v}|^2$

Spatial derivative order

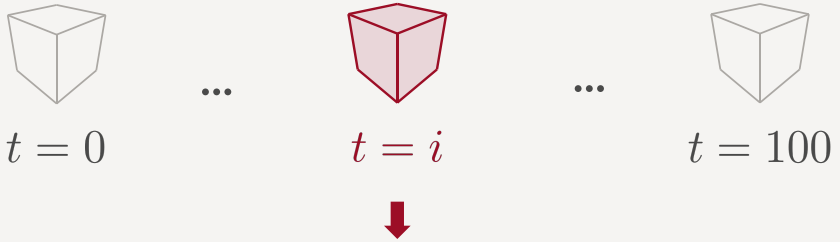
$\nabla\mathbf{A}, \quad \nabla^2\mathbf{A}$

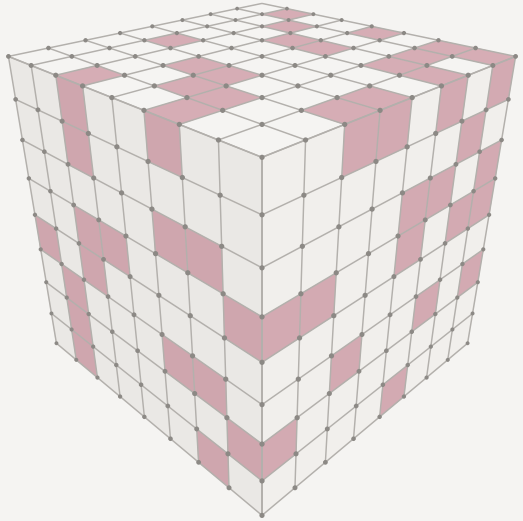
Number of summed terms

$\phi_1 + \phi_2, \quad \phi_1 - \phi_2$

Simulation parameters and acquisition cost

512^3 points · 101 snapshots · 7 components $(\rho, \mathbf{A}, \mathbf{v})$





512 blocks · 64^3 points per block

Acquisition A: read $m = 128$ blocks.
Evaluate all unresolved candidates.

Modeled cost per acquisition

$$C(\mathcal{U}, A) = m(C_{\mathsf{I/O}} + C_{\mathsf{eval}}|\mathcal{U}|)$$

m	blocks read
$\lvert\mathcal{U}\rvert$	unresolved candidates
$C_{\mathsf{I/O}}$	read cost per block
C_{eval}	cost per candidate per block

Algorithmic overhead is measured separately.

Defining invariance

We cannot establish a physical conservation law just by reading data. Instead, we test approximate invariance over the saved trajectory. A candidate is “invariant” if:

$$q_{\text{rel}} = \frac{\sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(I(t) - \bar{I}\right)^2}}{|\bar{I}|} \leq \epsilon_{\text{rel}}$$

(Temporal standard deviation over magnitude of temporal mean)

$I(t) = \int_{\Omega} \phi(x,t)\, d^3x$ — Integrate the candidate expression over the domain.

$\bar{I} = \frac{1}{T}\sum_{t=1}^{T} I(t)$ — Temporal mean across all timesteps in the trajectory.

ϵ_{rel} — A chosen tolerance, e.g. 0.05 or 0.10.

Algorithm sketch

Assumptions: Fixed block count at every read; one new timestep per read. Candidates modeled independently. Preset limits on cost and total reads.

```
Initialize        Read three pilot timesteps; initialize candidate models.
while unresolved candidates remain and another read fits the limits:
    Acquire       Choose the next timestep to read.
    Read          Sample blocks; evaluate every unresolved candidate.
    Update        Update each candidate model using the new observations.
    Classify      Label each candidate invariant, not invariant, or unresolved.
    Retire        Stop evaluating resolved candidates.
return labels, including any still unresolved.
```

The main methodological choices are how to do **acquisition** and **classification**.

Trajectory model and observation uncertainty

For each candidate,

$$\hat{I}(t) = \underbrace{\beta_0}_{\text{level}} + \underbrace{\beta_1 \tilde{t}}_{\text{drift}} + \underbrace{f(t)}_{\substack{\text{trajectory} \\ \text{variation}}} + \underbrace{\varepsilon_{\text{obs},t}}_{\substack{\text{sampling} \\ \text{noise}}}$$

Time-correlated trajectory variation

$$f \sim \mathcal{GP}\left(0, \sigma^2 \tilde{k}_\ell\right)$$

$$k_\ell(t, t') = \exp\left[-\frac{(\tilde{t} - \tilde{t}')^2}{2\ell^2}\right]$$

The trajectory is modeled as a Gaussian process. Tildes indicate that time is centered/scaled and the kernel projects out level and drift.

Sampling uncertainty

$$\hat{I}(t) = \frac{M}{m} \sum_{b \in A_t} I_b(t)$$

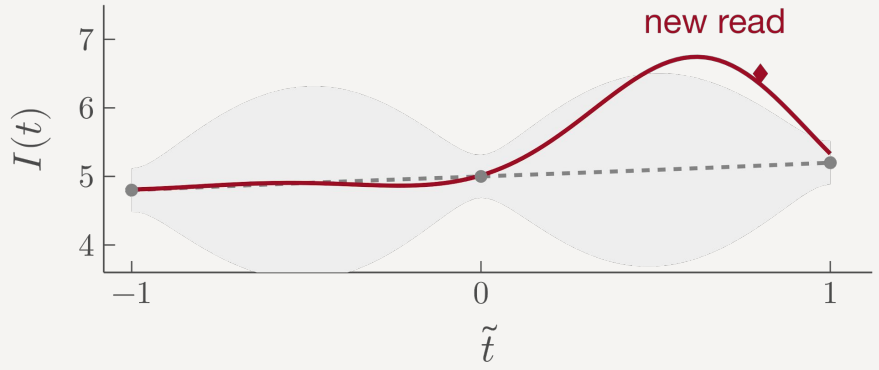
m of M blocks

$$\varepsilon_{\text{obs},t} \sim \mathcal{N}(0, s_t^2), \quad s_t^2 = M^2\left(1 - \frac{m}{M}\right)\frac{S_t^2}{m}$$

S_t^2 sample variance of block integrals

Initialization and posterior update

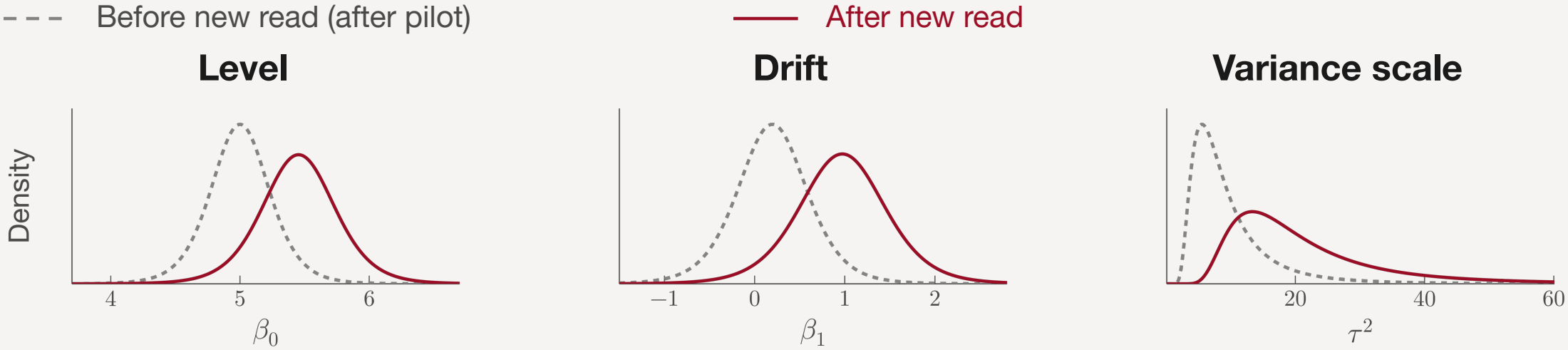
Illustrative example: the model is initialized from three pilot reads.



The new read is higher than predicted, raising the inferred level and drift. The surprise also increases the shared variance scale.

Priors: $\beta_0 \mid \tau^2 \sim \mathcal{N}(\hat{\beta}_0, \tau^2), \quad \beta_1 \mid \tau^2 \sim \mathcal{N}(0, \tau^2)$

$\tau^2 \sim \text{InverseGamma}(2, \hat{\beta}_0^2)$



Acquisition: D-optimal criteria

Choose the read with the greatest expected entropy reduction of model parameters, summed over unresolved candidates. Closed-form under Gaussian moment matching.

D-optimal

$$\theta = (\beta_0/|\hat{\beta}_0|,\ \beta_1/|\hat{\beta}_0|,\ \log\psi)$$

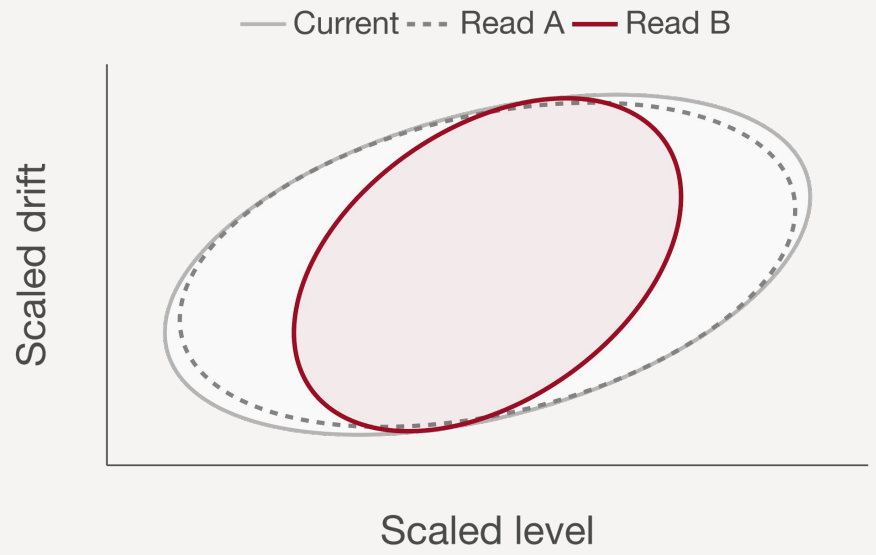
Learn the level, drift, and relative residual amplitude.

D_s-optimal

$$\theta_s = (\beta_1/|\hat{\beta}_0|,\ \log\psi)$$

Treat level as a nuisance; learn drift and residual amplitude.

$\psi = \sigma/\tau$ relative residual amplitude



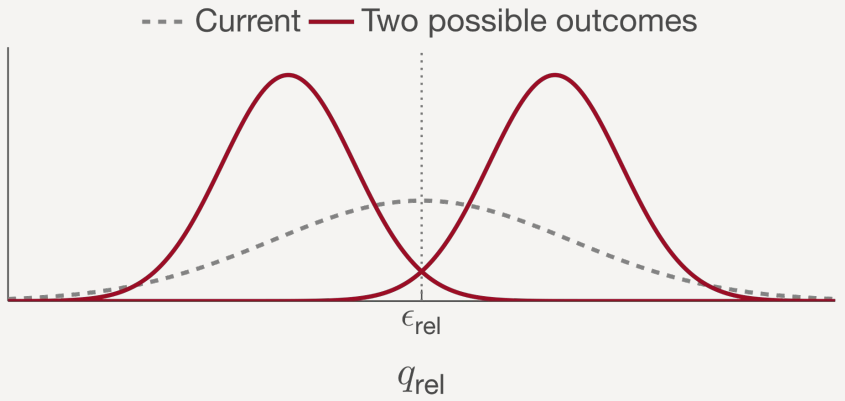
Select the read that shrinks uncertainty most.
In this example, Read B wins.

Acquisition: Delta-q and baselines

The Delta-q method targets the invariance decision directly, scoring reads by expected entropy reduction of the event:

$$\{q_{\text{rel}} \leq \epsilon_{\text{rel}}\}$$

We use a local linear approximation to predict how each read changes this decision, then choose the greatest expected information gain.



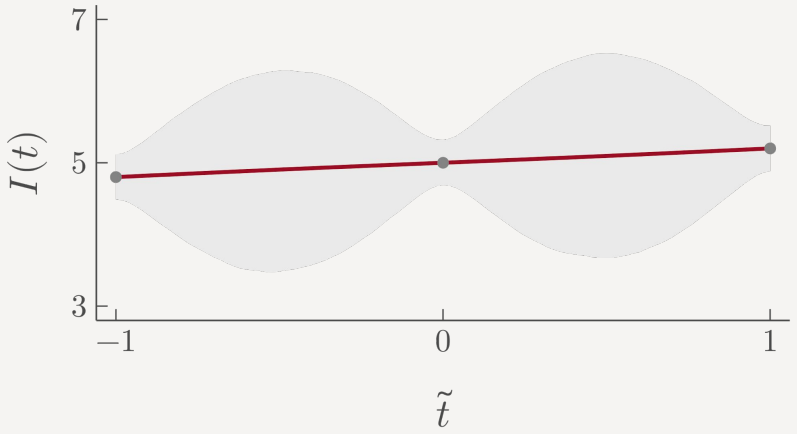
Acquisition baselines

LSE-Straddle	Read where the integral might be departing from its usual level, but the model is still too uncertain to tell.
Maximin	Read furthest from the nearest previously observed timestep.
Random	Choose an eligible unread timestep uniformly at random.

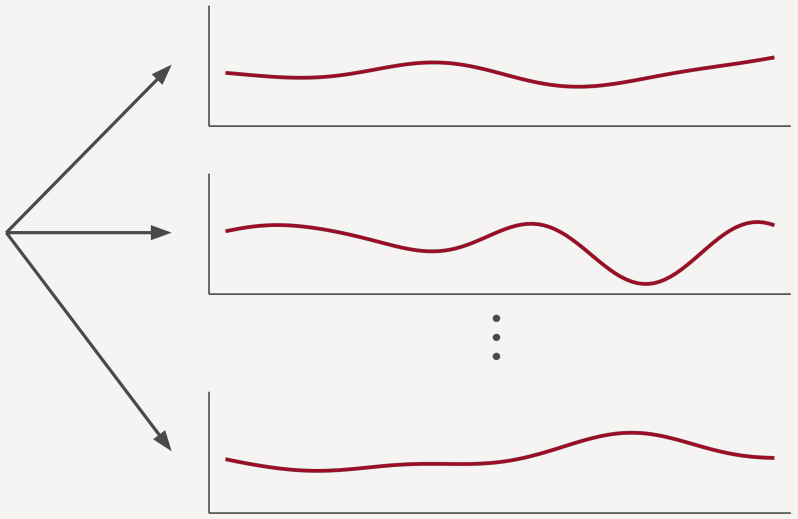
Classification: Bayes gate

Draw possible full trajectories from the posterior and test each one for invariance.

Posterior



Selected full-trajectory draws



$q_{\text{rel}}^{(1)} \le \epsilon_{\text{rel}}$

Passes the definition

$q_{\text{rel}}^{(2)} > \epsilon_{\text{rel}}$

Fails the definition

$q_{\text{rel}}^{(N)} \le \epsilon_{\text{rel}}$

Passes the definition

$$\hat{p}_{\text{inv}} = \frac{1}{N} \sum_{s=1}^{N} \mathbf{1}\left\{ q_{\text{rel}}^{(s)} \le \epsilon_{\text{rel}} \right\}$$

Calculate the fraction of draws that pass.

$\hat{p}_{\text{inv}} \ge 99\%$
Invariant

$\hat{p}_{\text{inv}} \le 5\%$
Not invariant

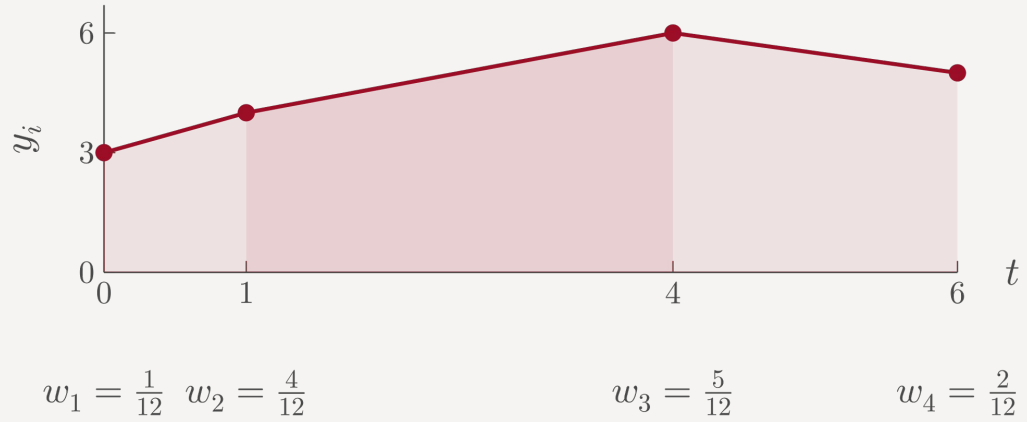
$5\% < \hat{p}_{\text{inv}} < 99\%$
Unresolved

Classification: trapezoid leave-one-out (LOO)

Estimate relative variation from the observed reads, weighting by the “trapezoid rule,” then check sensitivity to leaving each read out.

Weight by time coverage

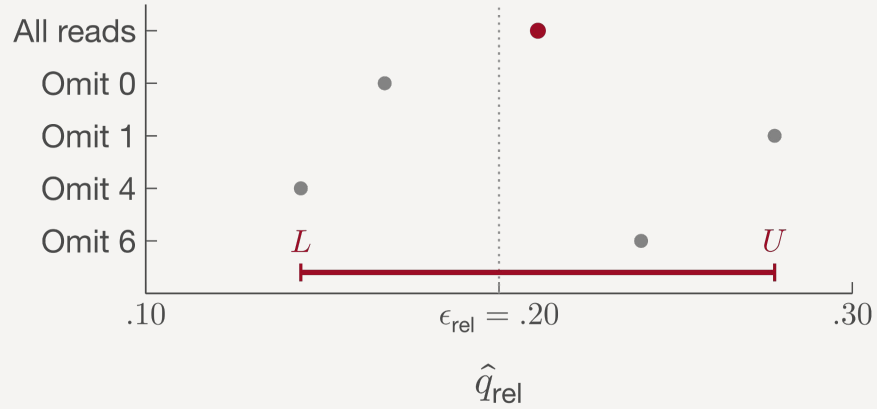
Each read shares half of each adjacent gap.



$$\hat{\mu} = \sum_i w_i y_i, \quad \sum_i w_i = 1$$

Leave one read out

Recompute the weights each time.



Take the min and max of all estimates, e.g.

$$[L, U] = [0.144,\ 0.278]$$

$U \leq \epsilon_{\text{rel}}$ **Invariant**

(Requires two confirmations)

$L > \epsilon_{\text{rel}}$ **Not invariant**

Otherwise: **Unresolved**

Benchmark design

$$\underbrace{556}_{\text{candidates}} \times \underbrace{20}_{\text{trajectories}} \times \underbrace{5}_{\text{tolerances}} = 55{,}600 \text{ total items}$$

Every method classifies the full bank; we focus evaluation on 601 hard items.

Budgets for each run

At most 32 frame reads, including the pilot.

$$C_{\max} = 8M(C_{\mathrm{I/O}} + C_{\mathrm{eval}}|\mathcal{C}|)$$

Equivalent to 8 full-resolution frames evaluating the entire candidate bank.

$$\epsilon_{\mathrm{rel}} \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$$

Hard set: 601 items

$$1.02 \le \max\left\{\frac{q_{\mathrm{rel}}}{\epsilon_{\mathrm{rel}}}, \frac{\epsilon_{\mathrm{rel}}}{q_{\mathrm{rel}}}\right\} \le 1.25$$

Multiplicative distance from the threshold



Empirically,
Below 2%: nearly impossible within the limits.
Above 25%: easy across tested methods.

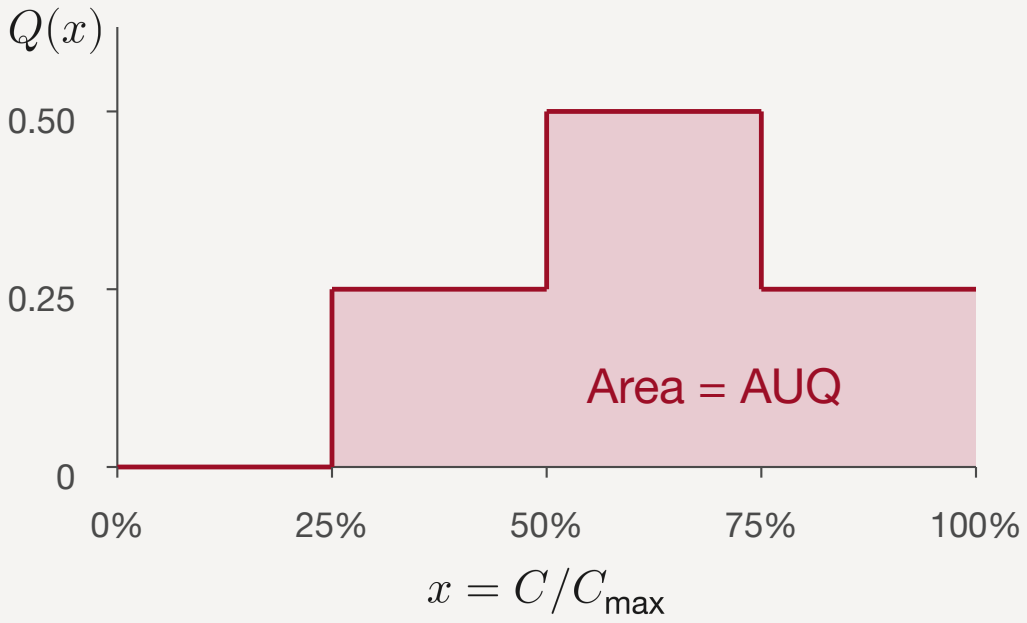
Main metric: area under quality (AUQ)

Reward early correct decisions, penalize early mistakes. Unresolved items contribute zero.

$$Q(x) = \frac{N_{\text{correct}}(x) - N_{\text{incorrect}}(x)}{N_{\text{items}}}$$

Budget used	Correct	Incorrect	U	Quality
0%	0	0	4	0.00
25%	1	0	3	0.25
50%	2	0	2	0.50
75%	2	1	1	0.25
100%	2	1	1	0.25

Area under the quality curve



$$\text{AUQ} = \int_0^1 Q(x)\,dx = \frac{1}{4}\left(0 + \frac{1}{4} + \frac{1}{2} + \frac{1}{4}\right) = 0.25$$

Main result: hard-set AUQ

601 hard items, pooled across runs; higher is better.

Acquisition	Trapezoid LOO			Bayes gate
	$m = 32$	$m = 128$	$m = 512$	$m = 128$
Random	0.409	0.562	0.345	0.480
Maximin	0.442	0.563	0.344	0.462
LSE-Straddle	0.383	0.494	0.359	0.437
D-optimal	0.434	**0.613**	0.406	0.552
D_s-optimal	0.405	0.575	0.406	0.433
Delta-q	0.401	0.560	0.381	0.496

- 128-block trapezoid LOO leads across all six acquisition methods.
- *D*-optimal is best (0.613); matched Bayes-gate setups score lower.

Final accuracy and computational tradeoffs

Overhead is wall-clock time spent choosing reads, updating models, and classifying, as a percentage of modeled read/evaluation cost.

Correct / incorrect / unresolved

Acquisition	Trapezoid LOO			Bayes
	$m = 32$	$m = 128$	$m = 512$	$m = 128$
Random	404/150/47	473/84/44	469/26/106	428/60/113
Maximin	410/135/56	452/62/87	471/25/105	416/63/122
LSE-Straddle	414/176/11	464/122/15	529/71/1	393/58/150
D-optimal	425/155/21	501/76/24	551/26/24	460/39/102
D_s-optimal	418/166/17	489/91/21	551/29/21	412/82/107
Delta-q	420/171/10	486/98/17	520/33/48	432/54/115

Computational overhead

Acquisition	Trapezoid LOO			Bayes
	$m = 32$	$m = 128$	$m = 512$	$m = 128$
Random	≈0%	≈0%	≈0%	28.9%
Maximin	≈0%	≈0%	≈0%	28.0%
LSE-Straddle	38.0%	12.1%	0.8%	69.3%
D-optimal	40.8%	11.9%	1.1%	116.3%
D_s-optimal	40.0%	11.6%	1.0%	101.7%
Delta-q	41.1%	12.3%	0.9%	85.8%

- *D*-optimal + full-resolution LOO ends with 551/601 correct, beating 128-block methods.
- Bayes gate and lower-resolution acquisition are computationally expensive.

Limitations and next steps

Classification	Better align classification with acquisition. Trapezoid LOO is heuristic, while Bayes gate is sensitive to priors and slow to resolve.
Adaptive resolution	Tested adaptive policies overuse coarse reads, incur high overhead, and have yet to outperform fixed resolution.
Physical invariance	Test a power-law decay criterion using a log–log model. This would change about 20–25% of hard-set labels, mostly toward invariant.
More trajectories	Current results use 20 of 60 generated trajectories. Evaluate the selected methods on the remaining 40 trajectories.